

Final Project Report

Care Seekers Healthcare



Project Supervisor

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Submitted By

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CERTIFICATE

This is to certify that [Muhammad Ahmad \(bc190405120\)](#), have worked on and completed their Software Project at Software & Research Projects Section, Department of Computer Sciences, Virtual University of Pakistan in partial fulfillment of the requirement for the degree of BS in Computer Sciences under my guidance and supervision.

In our opinion, it is satisfactory and up to the mark and therefore fulfills the requirements of BS in Computer Sciences.

Supervisor / Internal Examiner

[Faizan Tahir](#)

Supervisor,
Software Projects & Research Section,
Department of Computer Sciences
Virtual University of Pakistan

EXORDIUM

*In the name of Allah, the Compassionate, the
Merciful.*

*Praise be to Allah, Lord of Creation,
The Compassionate, the Merciful,
King of Judgment-day!*

*You alone we worship, and to You alone we pray
for help,
Guide us to the straight path*

The path of those who You have favored,

*Not of those who have incurred Your wrath,
Nor of those who have gone astray.*

DEDICATION

I dedicate this project to my mother, with great gratitude and immense love. Her support to me has been the greatest gift, and I hope to always make her proud.

ACKNOWLEDGEMENT

First and foremost, I extend my thanks to the Virtual University of Pakistan for providing me with the opportunity and for equipping me with the knowledge and resources necessary for its completion.

Furthermore, I am grateful to my project supervisor, Faizan Tahir, for guiding me and giving me advice throughout this project. His help has been really important.

PREFACE

The Care Seekers Healthcare project has taught me a lot about developing websites, especially using PHP. I discovered many new and interesting ways that websites work during my research.

It was really great working on this project and learning about web development. I am thankful to the Virtual University of Pakistan for giving me the opportunity to do this project.

I hope this project shows how much I've learned and inspires others to explore web development.

TABLE OF CONTENTS

Catalog

1. GATHERING & ANALYZING INFO	9
1.1 SCOPE	9
1.1.1 Modules	9
1.2 PROJECT REQUIREMENTS	9
1.2.1 Functional Requirements	9
1.2.2 Non-functional Requirements :	10
1.3 CASE DIAGRAM	11
1.4 USAGE SCENARIO	11
1.4.1 User Registration and Profile Management	11
1.4.2 Job Posting	12
1.4.3 Messaging Service	12
1.4.4 Job Application and Response	133
1.4.5 Job Acceptance	133
1.5 PROJECT METHODOLOGY	14
1.5.1 Waterfall methodology	14
1.5.2 Spiral methodology	15
1.5.3 Adopted methodology	15
1. 6 WORK PLAN	166
DESIGNING THE PROJECT	18
2.1 INTRODUCTION	18
2.2 ENTITY RELATIONSHIP DIAGRAM (ERD)	18
2.3 SEQUENCE DIAGRAM	19
2.3.1 Registration Sequence	19
2.3.2 Login Sequence	19
2.3.3 Job Posting Sequence	20
2.3.5 Messaging Sequence	20
2.4 Architecture Design Diagram	21
2.5 Class Diagram	21
2.6 Database Diagram	222
2.7 Interface Diagram	233
3. DEVELOPMENT	26
3.1 DEVELOPMENT PLAN	26
3.2 COMPLETE SOURCE CODE	266
3.3 DATABASE	266

CHAPTER 1

Gathering & Analyzing Info

1. Gathering & Analyzing Info

1.1 Scope

The objective of the project is to create a website focused on services, functioning as a platform that connects individuals seeking care with Support Workers in different fields including elder care, sick care, baby care, cooking, personal care, animal care, gym instruction, and domestic assistance. This web app will establish a community where Care Seekers can either publish job listings. Support Workers will register on the website and establish their hourly rates. Whenever a Care Seekers publishes a job, Support Workers will have the option to respond to it using a messaging service provided by the website. Upon finalizing the terms, the Support Worker can accept the job. The primary goal of the website is to simplify the process of finding and hiring Support Workers for a variety of care-related services.

1.1.1 Modules

This system mainly includes the following modules.

1. User Registration Module
2. Job Posting Module
3. Messaging Module
4. Job Acceptance Module
5. Profile Management Module
6. Review and Rating Module
7. Authentication and Security Module

1.2 Project Requirements

1.2.1 Functional Requirements

Care Seekers Functional Requirements:

1. User Registration: Care Seekers should be able to create an account and provide their necessary details, such as name, contact information, and address.
2. Job Posting: Care Seekers should have the ability to post job listings specifying the type of service required, service details, hourly budget, and preferred time.
3. Messaging Service: Care Seekers should be able to communicate with Support Workers through a messaging service to discuss job details, negotiate terms, and finalize agreements.
4. Job Selection: Care Seekers should have the ability to review Support Worker profiles and select the most suitable candidate for their job.
5. Rating and Reviews: Care Seekers should be able to provide ratings and reviews for the Support Workers they have hired, helping other Care Seekers make informed decisions.

Support Worker Functional Requirements:

1. **User Registration:** Support Workers should be able to register on the web app by providing their bio-data, picture, hourly rate, experience, and references.
2. **Job Application:** Support Workers should be able to browse and apply for available jobs posted by Care Seekers, expressing their interest and providing additional information if required.
3. **Messaging Service:** Support Workers should be able to communicate with Care Seekers through a messaging service to discuss job requirements, negotiate terms, and finalize agreements.
4. **Job Acceptance:** Support Workers should have the ability to accept jobs posted by Care Seekers, confirming their availability and commitment to providing the service.

Admin Functional Requirements:

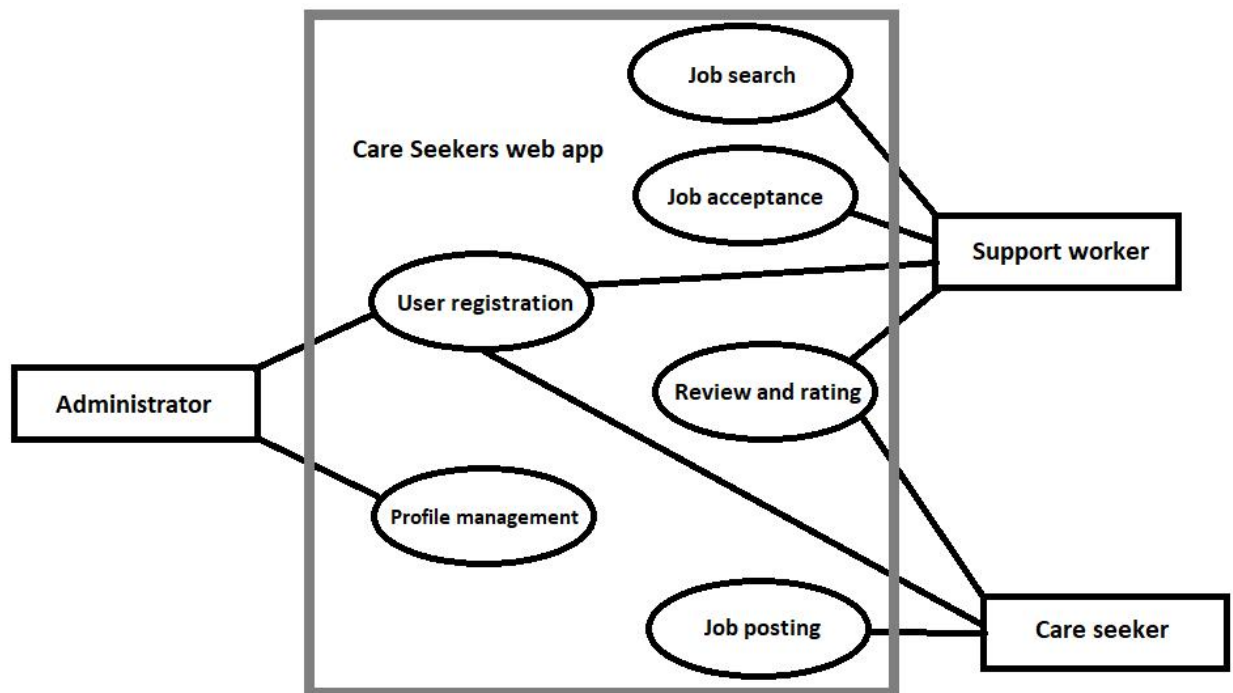
1. **User Management:** The admin should have the ability to manage user accounts, verify Support Worker details, and handle any reported issues or disputes.

1.2.2 Non-functional Requirements :

1. **Usability:** The web app should have an intuitive and user-friendly interface. The home page design must include a header with menus such as "Post Job," "Find Worker," "Testimonials," and "About". A footer should also be present on all pages.
2. **Design and Presentation:** The home page should have a visually appealing layout. It must include a centered section titled "Meet our Care and Support Workers" that showcases pictures and provides brief information about the workers.
3. **Security:** It should employ appropriate security measures, such as encryption and secure authentication protocols, to safeguard user information, including personal details and communication exchanged between Care Seekers and Support Workers.
4. **Scalability:** The app should be designed to handle an increasing number of users and data without compromising performance. As the platform grows, it should seamlessly accommodate new job postings, Support Workers, and user interactions.
5. **Reliability:** The web app should minimize downtime and disruptions, ensuring that Care Seekers and Support Workers can access the platform consistently and perform their desired actions without interruptions.
6. **Performance:** The app should load quickly, provide fast response times, and maintain smooth navigation throughout the site, even when faced with high user traffic. The images and content on the home page should be optimized.

By considering these non-functional requirements, the web app will not only fulfill the functional needs but also offer an aesthetically pleasing, secure, and user-friendly experience for Care Seekers, Support Workers, and the admin.

1.3 Case Diagram



1.4 Usage Scenario

The elaborated use cases for the Care Seekers modules, including the implementation priority, actors, summary, preconditions, normal course of events, alternative paths, and exceptions:

1.4.1 User Registration and Profile Management

- Implementation Priority: 1
- Actors: Care Seekers, Support Worker
- Summary: This use case allows Care Seekers and Support Workers to register on the web app and manage their profiles.
- Preconditions: The user must have access to the web app.

Normal Course of Events:

- The user selects the registration option.
 - The user provides the necessary details such as name, contact information, address, hourly rate, and experience.
 - The user submits the registration form.
 - The system validates the entered information and creates the user account.
- Exceptions: If the user enters invalid information or the system encounters an error, an appropriate error message is displayed and process is halted.

1.4.2 Job Posting

- Implementation Priority: 2
- Actors: Care Seekers
- Summary: Care Seekers can post job listings to find Support Workers.
- Preconditions: The Care Seekers must have an registered account.

Normal Course of Events:

- The Care Seekers selects the option to post a job.
 - The Care Seekers provides the job details (required service, service details, address, estimated hourly budget, and time of service).
 - The Care Seekers submits the job posting.
 - The system validates the entered information and publishes the job listing.
- Exceptions: If the entered information is incomplete or invalid, the system prompts the Care Seekers to correct the details.

1.4.3 Messaging Service

- Implementation Priority: 2
- Actors: Care Seekers, Support Worker
- Summary: Care Seekers and Support Workers can communicate with each other through a messaging service.
- Preconditions: The Care Seekers and Support Worker must be logged in and have active accounts. They must be engaged in a job posting or application process.

Normal Course of Events:

- The Care Seekers or Support Worker selects the messaging option.
 - The user navigates to the relevant conversation or starts a new conversation.
 - The user composes and sends a message.
 - The recipient receives and can respond to the message.
- Exceptions: If there are any technical issues with sending or receiving messages, an error message is displayed, and the user is informed about the failure.

1.4.4 Job Application and Response

- Implementation Priority: 2
- Actors: Support Worker
- Summary: Support Workers can browse and apply for job listings posted by Care Seekers, and Care Seekers can respond to Support Worker applications.
- Preconditions: The Support Worker must be logged in and have an active account. There must be available job listings.

Normal Course of Events:

- The Support Worker selects the option to browse job listings.
 - The Support Worker views available job listings and selects the desired job to apply for.
 - The Support Worker submits the application, providing any additional information if required.
 - The Care Seekers reviews the application and can accept or reject it.
- Alternative Path: If there are no available job listings or if the Support Worker's application is rejected, the system notifies the Support Worker accordingly.
 - Exceptions: If there are any technical issues with the application or response process, an appropriate error message is displayed, and the user is informed about the failure.

1.4.5 Job Acceptance

- Implementation Priority: 2
- Actors: Support Worker
- Summary: Support Workers can accept job offers from Care Seekers, confirming their commitment to providing the service.
- Preconditions: The Support Worker must be logged in and have an active account. The Support Worker should have received a job offer.

Normal Course of Events:

- The Support Worker views the job offer notification.
 - The Support Worker accepts the job offer, indicating their commitment.
 - The Care Seekers is notified of the Support Worker's acceptance.
- Alternative Path: If the Support Worker declines the job offer or fails to respond within a specified time frame, the Care Seekers is notified accordingly.

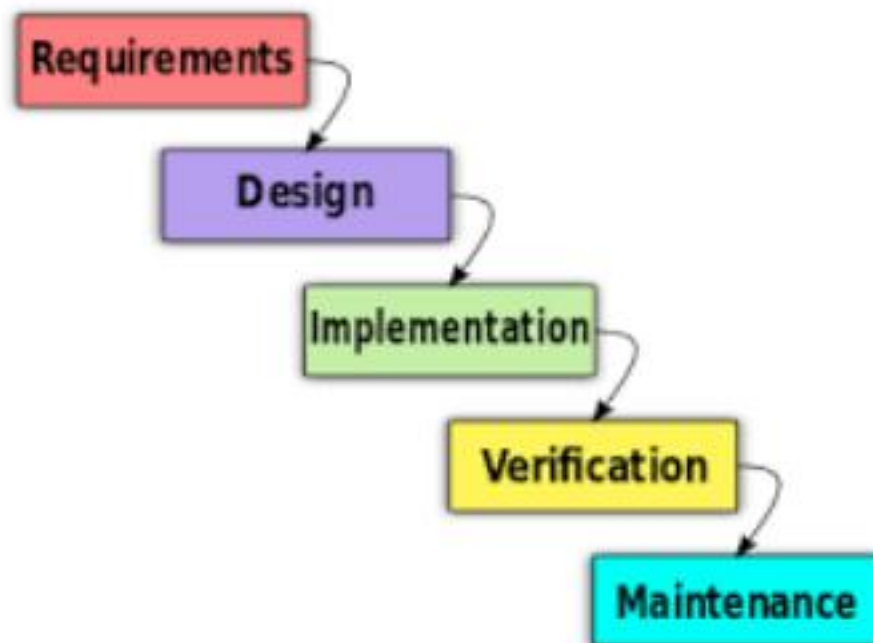
- Exceptions: If there are any technical issues with the job acceptance process, an appropriate error message is displayed, and the user is informed about the failure.

These use cases provide a comprehensive overview of the main functionalities of the web app, allowing for smooth interactions and effective management of job postings, communication, and acceptance between Care Seekers and Support Workers.

1.5 Project Methodology

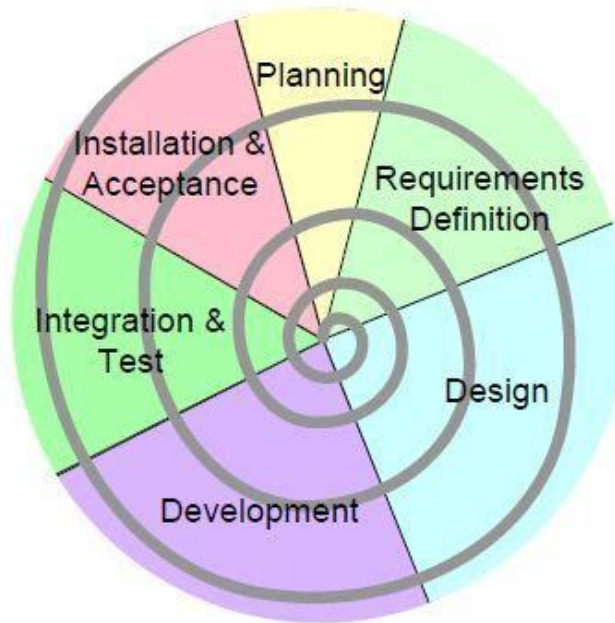
1.5.1 Waterfall methodology

The Waterfall model is a linear approach to software development that follows a sequential path from conception to maintenance. It is characterized by a steady progression through phases such as initiation, analysis, design, construction, testing, production/implementation, and maintenance. Despite the emergence of alternative software development models, the Waterfall method remains widely adopted, with more than a third of software developers still relying on it as the primary approach.



1.5.2 Spiral methodology

The spiral model is a risk-driven process model generator for software projects. Based on the unique risk patterns of a given project, the spiral model guides a team to adopt elements of one or more process models, such as incremental, waterfall, or evolutionary prototyping.



1.5.3 Adopted methodology

My adopted methodology is combining the Waterfall and Spiral development methodologies. The VU Process methodology combines the structured nature of Waterfall with the risk management aspects of Spiral.

The Care Seekers project will be developed using VU Process model. The project will begin with the initial requirements phase, after that we will use the cyclic nature of Spiral model to design, test, develop and repeat. The project progresses through these cycles, allowing for incremental development and improvement.

A few reasons for using this model for my project :

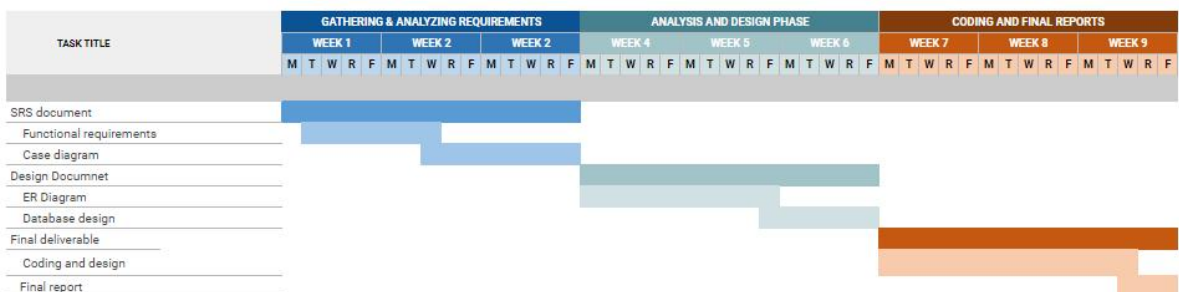
First of all, it is encouraged by the University that VU Process model be used. I will use Waterfall nature of VU Process model for solid development and design. The Spiral component of VU Process will help with risk control and repeat improvement as the project progresses towards the final deliverable.

1. 6 Work Plan

All projects of any nature must have clear and solid work plan if they are to succeed. Following is the Gantt chart of the project.

Care Seeker Project timeline

PROJECT TITLE Care Seekers
PROJECT SUPERVISOR Faizan Tahir



CHAPTER 2

Designing the Project

2. Designing the Project

2.1 Introduction

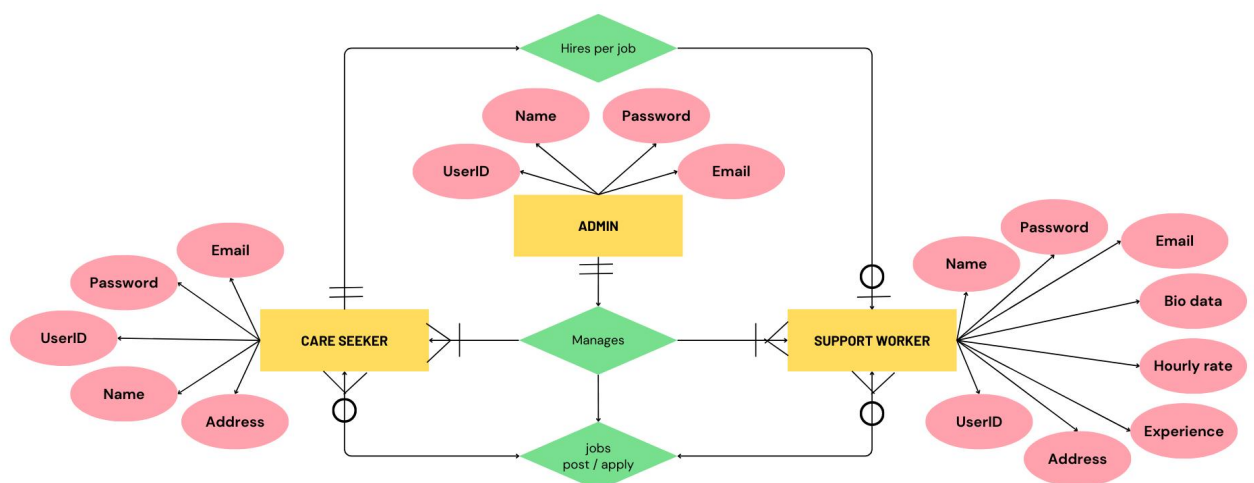
The Design Document describes the whole information about the software. During the design phase, the focal point shifts from what to how.

A design document is like a big plan for solving a problem. It's really important to make a system. Making a plan for the software is all about breaking down the system into smaller parts to get the result you want. It's like drawing a map to build something by explaining how the system should look when it's done.

The objective of the design process is to analyze and understand the system in detail so that features and constituent components of at least one feasible solution are identified and documented. The design activity provides a road map to progressively transform the requirements through a number of stages into the final product by describing the structure of the system to be implemented.¹

Entity Relationship diagram, sequence diagrams, class diagram, interface diagrams, test cases as well as database and architecture designs are built-in in this design document.

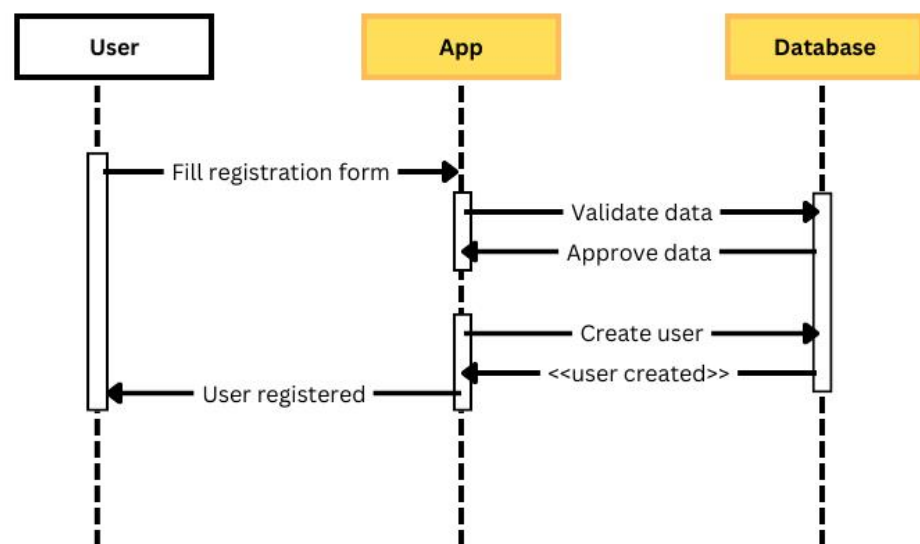
2.2 Entity Relationship Diagram (ERD)



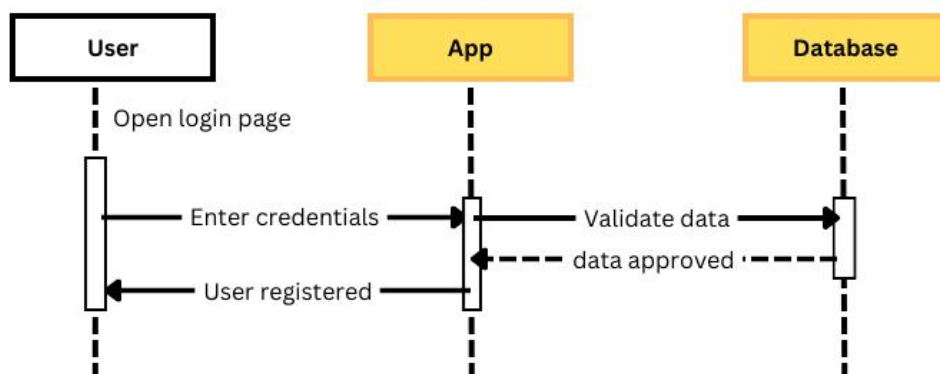
2.3 Sequence Diagram

The purpose of sequence diagrams is to show how different parts of a system or software interact with each other over time. They help in visualizing the flow of actions and messages between different components or objects within the system. Sequence diagrams are used to understand and communicate the dynamic behavior of a system, including the order in which actions or events occur and how they affect the system's functionality. Essentially, sequence diagrams provide a clear and detailed picture of how the system's components work together to achieve specific tasks or processes.

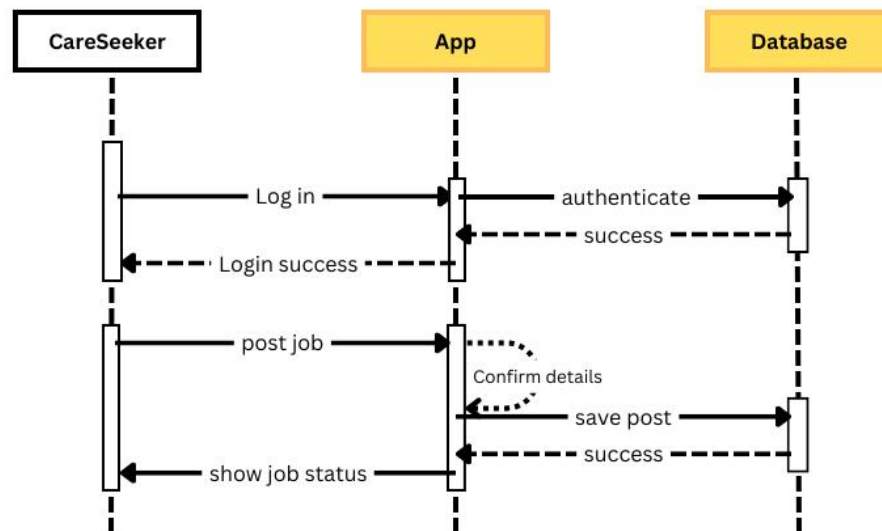
2.3.1 Registration Sequence



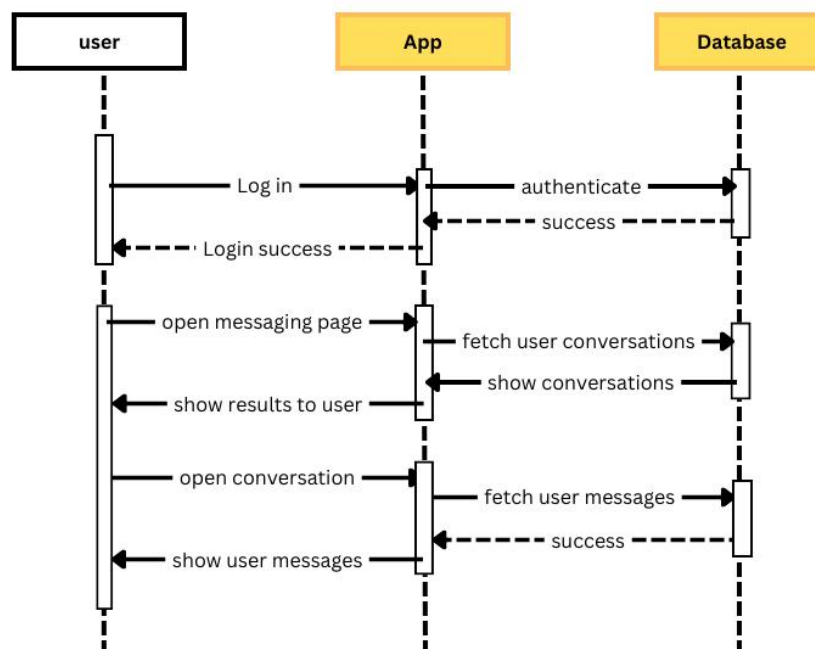
2.3.2 Login Sequence



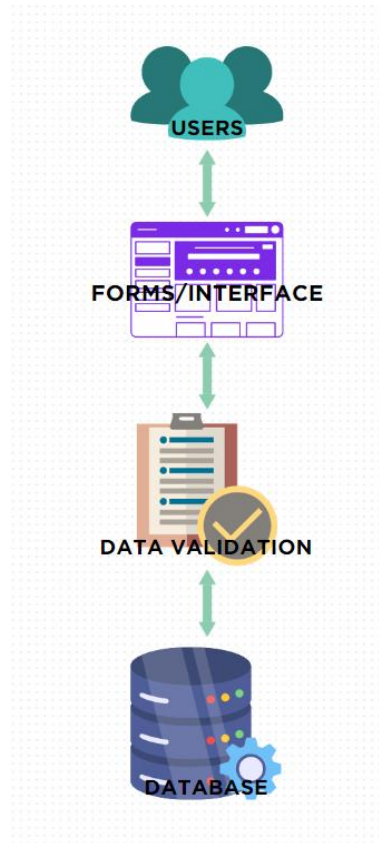
2.3.3 Job Posting Sequence



2.3.4 Messaging Sequence



2.4 Architecture Design Diagram

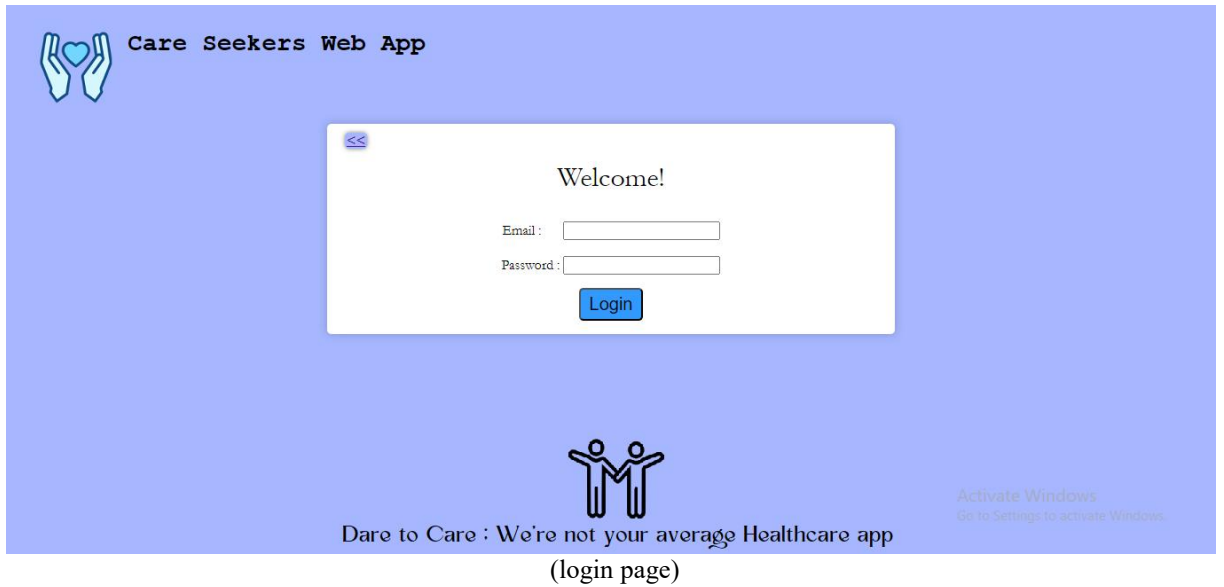


2.5 Class Diagram

A class diagram is a visual representation in software design that shows the structure and relationships of classes in a system or program. Classes are like blueprints for objects in the software, defining their attributes (data) and behaviors (functions or methods). In a class diagram:

1. **Classes** : Rectangles represent classes and contain the name of the class at the top.
2. **Attributes** : Inside a class, attributes (variables or properties) are listed along with their data types. These represent the data the class holds.
3. **Methods** : Inside a class methods (functions or behaviors) are also listed, showing what actions can be performed.
4. **Arrows** : Lines with arrows connecting classes indicate relationships or associations between them. For example, a line between two classes might represent that one class uses or depends on the other.

2.7 Interface Diagram



The login page features a light blue background. At the top left is a logo of two hands holding a heart, followed by the text "Care Seekers Web App". In the center is a white login box with a back arrow icon in the top left corner. Inside the box, it says "Welcome!" followed by input fields for "Email:" and "Password:", and a blue "Login" button. Below the login box is an icon of two people holding hands, with the text "Dare to Care : We're not your average Healthcare app" and "(login page)". In the bottom right corner, there is a watermark that says "Activate Windows Go to Settings to activate Windows."

Care Seekers Web App

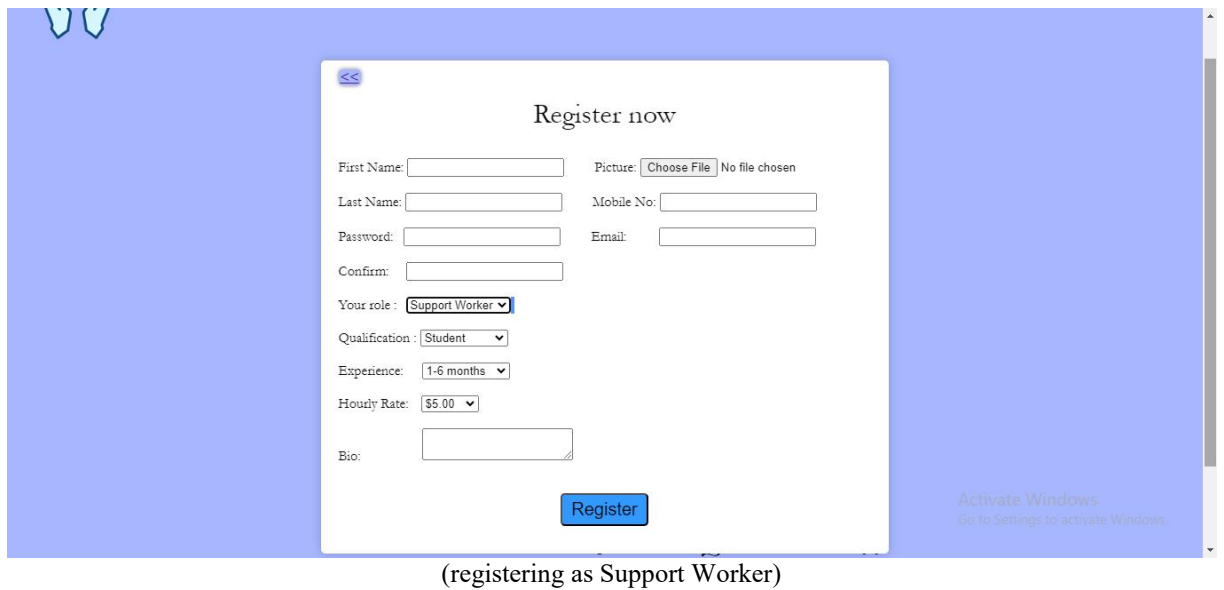
Welcome!

Email :

Password :

Login

Dare to Care : We're not your average Healthcare app
(login page)



The register page has a light blue background. At the top left is a logo of two hands holding a heart. In the center is a white registration box with a back arrow icon in the top left corner. Inside the box, it says "Register now" followed by input fields for "First Name:", "Last Name:", "Mobile No:", "Password:", "Email:", and "Confirm:". There are also dropdown menus for "Your role:" (selected "Support Worker"), "Qualification:" (selected "Student"), "Experience:" (selected "1-6 months"), and "Hourly Rate:" (selected "\$5.00"). A "Picture:" field with a "Choose File" button and "No file chosen" text is also present. A "Bio:" field with a text area is at the bottom. A blue "Register" button is at the bottom right of the box. In the bottom right corner of the page, there is a watermark that says "Activate Windows Go to Settings to activate Windows."

Register now

First Name: Picture: No file chosen

Last Name: Mobile No:

Password: Email:

Confirm:

Your role:

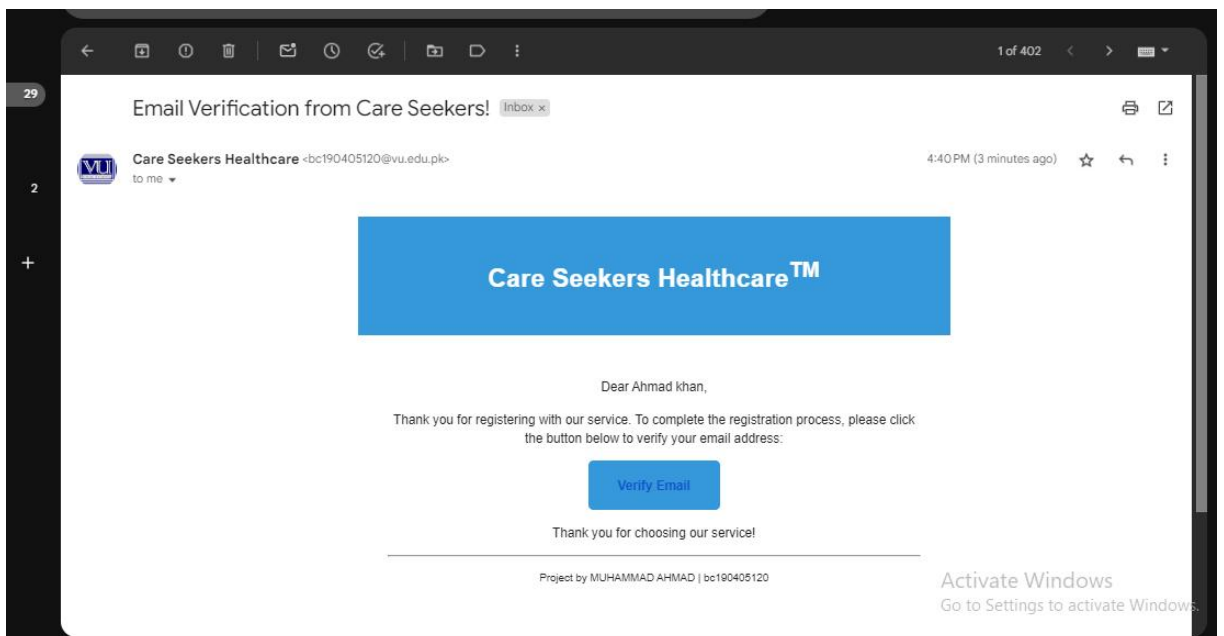
Qualification:

Experience:

Hourly Rate:

Bio:

Register



The email verification page is shown within a browser window. The email header says "Email Verification from Care Seekers!" and "Inbox x". The sender is "Care Seekers Healthcare" with the email address "<bc190405120@vu.edu.pk>". The subject is "to me". The body of the email features a blue banner with "Care Seekers Healthcare™". Below the banner, it says "Dear Ahmad Khan," followed by "Thank you for registering with our service. To complete the registration process, please click the button below to verify your email address:". A blue "Verify Email" button is centered. Below the button, it says "Thank you for choosing our service!". At the bottom, it says "Project by MUHAMMAD AHMAD | bc190405120". In the bottom right corner of the browser window, there is a watermark that says "Activate Windows Go to Settings to activate Windows."

Email Verification from Care Seekers! Inbox x

Care Seekers Healthcare <bc190405120@vu.edu.pk>
to me

4:40 PM (3 minutes ago) ☆ ↶ ⋮

Care Seekers Healthcare™

Dear Ahmad Khan,

Thank you for registering with our service. To complete the registration process, please click the button below to verify your email address:

Verify Email

Thank you for choosing our service!

Project by MUHAMMAD AHMAD | bc190405120

(Email verification)

**Care Seekers Healthcare**

Admin Panel

Hide Applications¹

About

Log Out

Picture	General Information	Job Information	Bio	Approval
	Full name : Ahmad Khan Email : theelderking7337@gmail.com Phone # : 01001122333	Role : seeker Address : abc address	abc address	Accept Reject



Dare to Care : We're not your average Healthcare app

Activate Windows
Go to Settings to activate Windows.

(Admin approval)

CHAPTER 3

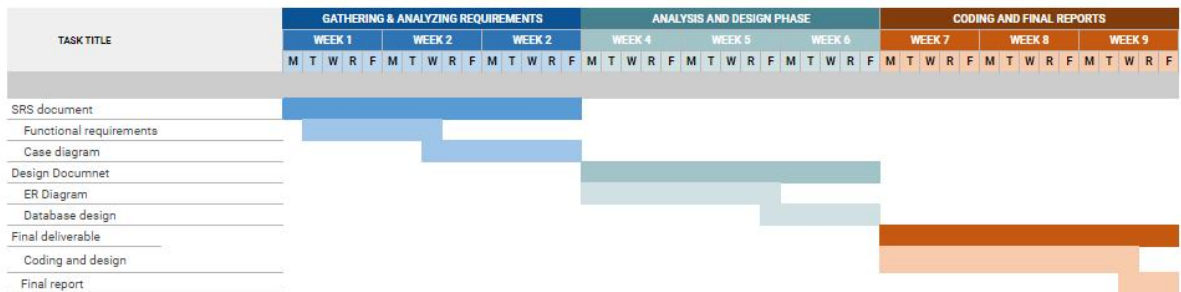
Development

3. Development

3.1 Development Plan

Care Seeker Project timeline

PROJECT TITLE Care Seekers
PROJECT SUPREVISOR Faizan Tahir



3.2 Complete Source Code

The source code for the complete is provided along with this report in its entirety.

3.3 Database

The the database for this project is provided along with this report.